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Hybrid optimization schemes for quantum control

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Abstract

Optimal control theory is a powerful tool for solving control problems in quantum mechanics, ranging from the control of chemical reactions to the implementation of gates in a quantum computer. Gradient-based optimization methods are able to find high-fidelity controls, but require considerable numerical effort and often yield highly complex solutions. We propose here to employ a two-stage optimization scheme to significantly speed up convergence and achieve smoother controls. The control is initially performed using only a few parameters of a guess pulse, before switching to a gradient-based algorithm in the second stage. We illustrate the performance of the new scheme by comparing it to standard gradient-based optimization for the robust preparation of a Bose-Einstein condensate in a one-dimensional optical lattice, and for the implementation of a two-qubit phase gate for two coupled Rydberg atoms. Both test cases are known to exhibit slow convergence due to multiple traps. We find that the new scheme is able to speed up the optimization by an order of magnitude, while yielding smoother controls. Moreover, it is shown to be more robust against numerical noise, which is often a limiting factor for the success of gradient-based optimization.

Keywords: [illegible]

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1 Introduction

Coherent quantum control has long been a key component in the effort towards future quantum technologies. It relies on the interference between multiple pathways to steer a quantum system in some desired way [1, 2]. Originally conceived for the control of chemical reactions [3–5], it has since been extended to a wide variety of applications, see, e.g. [6] for a review. In the context of optimal control theory (OCT), a guess control is gradually altered in order to steer the dynamics in the optimization target. Given the fastest converging algorithms are those that take into account information about the gradient of the optimization functional with respect to variations in the control. Gradient based methods operate in some loop, where the entire optimization procedure is performed based on the knowledge of the system dynamics. Two widely used methods are known as method [7–9] and gradient ascent pulse engineering (GRAPE) [10]. Krotov's method guarantees monotonic convergence for time-continuous control fields. The Limited-memory Broyden–Fletcher–Goldfarb–Shanno (LBFGS) method [11] can be used in concert with GRAPE [12] and Krotov's method [13], considering not only the gradient but also an estimate for the Hessian, i.e., the second order derivatives. This has been demonstrated to

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